Chapter 3

Exercise 7.1. Assume that the complex

$$\ldots \longrightarrow 0 \longrightarrow M \longrightarrow 0 \longrightarrow \ldots$$

is exact. Prove $M \cong 0$.

Proof. Let φ be the map out of M; the complex being exact at M means φ is injective; there $M \cong 0$. □

Exercise 7.2. Assume that the complex

$$\ldots \longrightarrow 0 \longrightarrow M \longrightarrow M' \longrightarrow 0 \longrightarrow \ldots$$

is exact. Prove $M \cong M'$.

Proof. Let ψ be the map between M and M' and φ be the map out of M'. Since the complex is exact, $\operatorname{img}\psi \cong M/0$ and $\operatorname{img}\psi \cong \ker\varphi = M'$; thus $M \cong M'$.

$$\ldots \longrightarrow 0 \longrightarrow M \xrightarrow{\psi} M' \xrightarrow{\varphi} 0 \longrightarrow \ldots$$

□

Exercise 7.3. Assume that the complex

$$\ldots \longrightarrow 0 \longrightarrow L \longrightarrow M \xrightarrow{\varphi} M' \longrightarrow N \longrightarrow 0 \longrightarrow \ldots$$

is exact. Show that up to natural identification $L = \ker\varphi$ and $N = \operatorname{coker}\varphi$.

Proof. TODO

□

Chapter 6

Exercise 1.1. Show that $\mathbb{C}$ and $\mathbb{R}$ are isomorphic as $\mathbb{Q}$-vector spaces.

Proof. Both $\mathbb{C}$ and $\mathbb{R}$ are infinite dimensional vector spaces over $\mathbb{Q}$; any basis must be uncountable with cardinality $2^{\aleph_0}$ (otherwise we could express every real number as a finite sum of a countable basis elements and so $\mathbb{R}$ would be a countable set). Take any isomorphism ψ between $\mathbb{C}$ and $\mathbb{R}$ as sets with $\psi(0) = 0$ and $\psi(x) = -x$ - these exist but are non-constructible; then ψ maps the basis of $\mathbb{C}$ into the basis of $\mathbb{R}$ and they are isomorphic as $\mathbb{Q}$-modules. □

Exercise 1.2. Prove that the sets listed in Exercise III.1.4 are all $\mathbb{R}$-vector spaces, and compute their dimension.

1. $\mathfrak{sl}_n(\mathbb{R})$ - $n \times n$ matrices over $\mathbb{R}$ with trace 0
2. $\mathfrak{sl}_n(\mathbb{C})$ - $n \times n$ matrices over $\mathbb{C}$ with trace 0
3. $\mathfrak{so}_n(\mathbb{R})$ - $n \times n$ matrices over $\mathbb{R}$ with $MM^t = I$
4. $\mathfrak{su}(n)$ - $n \times n$ matrices over $\mathbb{C}$ with $MM^t = I$

Proof. TODO □

Exercise 1.5. Let R be an integral domain. Prove or disprove the following:

False: *Every linearly independent subset of a free R-module may be completed to a basis.*

Proof. Let $R = \mathbb{Z}$ and take the subset $\{2\}$; this cannot be completed to a basis. □

TODO

Exercise 1.13. Let A be an abelian group such that $\mathrm{End}_{Ab}(A)$ is a field of characteristic 0. Prove that $A \cong \mathbb{Q}$.

Proof. Since the endomorphism ring of A is a field, every endomorphism φ has an inverse φ^{-1}.

Every abelian group is a $\mathbb{Z}$-module via $na = \underbrace{n + \ldots + n}_{n \text{ times}}$. Call this φ_n; then φ_n is an endomorphism of A and so has an inverse $\varphi_{1/n}$. A then is a $\mathbb{Q}$ vector space by $\frac{p}{q}a = \varphi_p \varphi_{1/q} a$.

The dimension of this module must equal 1; otherwise we would have non-zero endomorphisms ψ of A with $\ker \psi = 0$ (set everything except the first dimension to 0). These endomorphisms would not have an inverse, contradicting the endomorphism ring being a field. Therefore A is a $\mathbb{Q}$ vector space with dimension 1 and so $A \cong \mathbb{Q}$. □

Exercise 1.14. Let V be a finite dimensional vector space and $\varphi : V \to V$; show that there exists an N such that $\ker \varphi^{n+1} = \ker \varphi^n$ and $\operatorname{img} \varphi^{n+1} = \operatorname{img} \varphi^n$.

Proof. Say V is a finite dimension vector space over a field k. We rely on the following:

Claim 1: Any two vector spaces of the same dimension over k are isomorphic.

Claim 2: Let V have finite dimension; then $\dim \varphi(V) \leq \dim V$.

Given these claims, applying φ to V either maintains the dimension of V (giving an isomorphic image and so $\ker \varphi = 0$) or strictly reduces the dimension of V, a non-negative integer. As V has finite dimension applying up to φ $\dim V$ times will stabilize the kernel to 0 and the image to some finite dimensional subspace of V. (Note the kernel is strictly 0 so that the images are strictly equal, vs just being isomorphic.)

Proof of Claim 1: Let V, V' be vector spaces over k with dimension n, with e_i be a basis for V and f_i be a basis for V'. Let $\psi : V \to V'$ be defined by $\psi(e_i) = f_i$; then ψ is surjective (since f_is are a basis of V'). Suppose $\psi(v) = 0$; then $\sum \psi(k_i e_i) = \sum k_i \psi(e_i) = \sum k_i f_i = 0$; as the f_i are linearly independent each of the k_is must be 0, therefore $v = 0$ and ψ is injective. Therefore ψ is an isomorphism.

Proof of Claim 2: A linearly dependent set of vectors is mapped by any endomorphism of vector spaces to a linearly dependent set of vectors, so the dimension under the mapping can't increase or become infinite.

We provide a counterexample for infinite-dimensional vector spaces. Take the vector space $k^{\oplus \mathbb{N}}$ and let $\varphi(\langle v_0, v_1, \ldots \rangle) = \langle v_1, v_2, \ldots \rangle$. Then any finite number of applications of φ has a non-trivial (but non-equal) kernel and image for any φ^k and φ^{k+1}. □

Exercise 1.15. Say R is an integral domain that is not a field and let F be a free module of finite rank. What property of R immediately guarantees $\ker \varphi^{n+1} = \ker \varphi^n$ for some $n > 0$?

Proof. **Claim:** If $\operatorname{rk} F = \operatorname{rk} \varphi(F)$, then $\ker \varphi = 0$.

Proof of Claim: Let $x \in \ker \varphi$ □

Show that there is an R-module homomorphism $\varphi : F \to F$ such that $\operatorname{img} \varphi^{n+1} \subsetneq \operatorname{img} \varphi^n$ for all $n > 0$.

Proof. □

Chapter 8

Exercise 2.1. Let M, N be R-modules and assume N is cyclic. Prove every element of $M \otimes_R N$ may be written as a pure tensor.

Proof. Every element of $M \otimes_R N$ has the form $\sigma_i r_i (m_i \otimes_R n_i)$ with $m_i \in M, n_i \in N$; since N is cyclic, every $n_i \in N$ can be written as $r_i' a$ for $a \in N$. Therefore

$$\begin{aligned} \sigma_i r_i (m_i \otimes_R n_i) &= \sigma_i r_i (m_i \otimes_R r_i' a) \\ &= \sigma_i r_i r_i' (m_i \otimes_R a) \\ &= (\sigma_i r_i r_i' m_i) \otimes_R a \end{aligned}$$

Since M is an R-module, each finite sum $r_i r_i' m_i \in M$, so every element of $M \otimes_R N$ is a pure tensor. □

Exercise 2.2. Prove "by hand" that $\mathbb{Z}/n\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/m\mathbb{Z} \cong 0$ if $gcd(m, n) = 1$.

Proof. For any pure tensor, $x \otimes 0 = x \otimes (0+0) = x \otimes 0 + x \otimes 0$, therefore $x \otimes 0 = 0$.

Since $gcd(m, n) = 1$, m has a multiplicative inverse in $\mathbb{Z}/n\mathbb{Z}$ and n has a multiplicative inverse in $\mathbb{Z}/m\mathbb{Z}$. Therefore given $x \in \mathbb{Z}/n\mathbb{Z}$, $y \in \mathbb{Z}/m\mathbb{Z}$:

$$\begin{aligned} x \otimes_{\mathbb{Z}} y &= (m \cdot m^{-1} \cdot x) \otimes_{\mathbb{Z}} (n \cdot n^{-1} \cdot y) \\ &= (n \cdot m^{-1} \cdot x) \otimes_{\mathbb{Z}} (m \cdot n^{-1} \cdot y) \\ &= (m^{-1} \cdot n \cdot x) \otimes_{\mathbb{Z}} (n^{-1} m \cdot \cdot y) \\ &= (m^{-1} \cdot 0) \otimes_{\mathbb{Z}} (n^{-1} \cdot 0) \\ &= 0 \otimes_{\mathbb{Z}} 0 = 0 \end{aligned}$$

All pure tensors are 0, therefore the tensor product is the 0 $\mathbb{Z}$-module. □

Exercise 2.3. Prove that

$$R[x_1, \ldots x_n] \otimes_R R[y_1, \ldots, y_n] \cong R[x_1, \ldots, x_n, y_1, \ldots, y_m]$$

Proof. Let $f \in R[x_1, \ldots x_n]$, $g \in R[y_1, \ldots y_m]$. Define $j : R[x_1, \ldots x_n] \otimes R[y_1, \ldots y_m] \to R[x_1, \ldots, x_n, y_1, \ldots, y_m]$ by $j(f, g) = fg$; we show j is bijective and surjective and so an isomorphism.

This map is well-defined:

$$j((x_i^k + f) \otimes g) = (x_i^k + f)(g) = x_i^k g + fg = j(x_i^k \otimes g) + j(f \otimes g) = j((x_i^k + f) \otimes g)$$

and similarly

$$j(f \otimes (y_i^k + g)) = fy_i^k + fg = j(f \otimes y_i^k) + j(f \otimes g) = j(f \otimes (y_i^k + g))$$

and is R-linear:

$$j(rf \otimes g) = rfg = rj(f \otimes g)$$

It is injective; if $fg = 0$ then consider $f \otimes g$; splitting the polynomials to reducing to monomials and grouping terms, $f \otimes g = \sum_i r_i(\prod_k x_k) \otimes (\prod_k y_k)$; if each monomial is 0 in the polynomial ring, each $r_i = 0$, so $f \otimes g = 0$.

It is surjective: given $h \in R[x_1, \ldots, x_n, y_1, \ldots, y_m]$, h can be written as the sum of monomials; each monomial is mapped to by the pure tensor pulling all x_is to the left and all y_js to the right. □

Exercise 2.4. Let S, T be commutative R-algebras. Verify the following:

TODO (straightforward)

Exercise 2.5. Let S be a multiplicative subset of R, and M be an R-module. Prove that $S^{-1}M \cong M \otimes_R S^{-1}R$ as R-modules.

Proof. Let $\psi : M \times S^{-1}R \to P$ be a bilinear map. Define $j : M \times S^{-1} \to S^{-1}M$ by $j(m, r/s) = mr/s$ and $\psi' : S^{-1}M \to P$ by $\psi'(m/s) = \psi(m, 1/s)$. ($\psi'$ is an R-module homomorphism by the bilinearity of ψ.) Then

$$\begin{aligned}(\psi' \circ j)(m, r/s) &= \psi'(mr/s) \\ &= \psi(mr, 1/s) \\ &= r\psi(m, 1/s) \\ &= \psi(m, r/s)\end{aligned}$$

By the universal property of the tensor product, $S^{-1}M \cong M \otimes_R S^{-1}R$.

□

Exercise 2.7. Changing the base ring in a tensor may or may not make a difference:

Prove that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q} \otimes_{\mathbb{Q}} \mathbb{Q}$.

Proof. Clearly a $\mathbb{Q}$-bilinear map is a $\mathbb{Z}$-bilinear map: move only the numerators.

For the other direction, let $\psi : \mathbb{Q} \times \mathbb{Q} \to P$ by a $\mathbb{Z}$-bilinear map; given $p \otimes q \in \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$,

$$(m/n)q \otimes_{\mathbb{Z}} p = (q/n) \otimes_{\mathbb{Z}} mp = (q/n) \otimes_{\mathbb{Z}} (n/n)(mp) = q \otimes_{\mathbb{Z}} (m/n)p$$

Therefore, ψ is a $\mathbb{Q}$-bilinear map and $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q} \otimes_{\mathbb{Q}} \mathbb{Q}$. □

Prove that $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \ncong \mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$.

Proof. $\mathbb{C} \cong \mathbb{R} \oplus \mathbb{R}$ (as a 2-dimensional vector space with scalars in $\mathbb{R}$); so

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \otimes_{\mathbb{R}} (\mathbb{R} \oplus \mathbb{R}) \cong (\mathbb{C} \otimes_R \mathbb{R}) \oplus (\mathbb{C} \otimes_R \mathbb{R}) \cong \mathbb{C} \oplus \mathbb{C} = \mathbb{C} \times \mathbb{C}$$

However, $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C} \cong \mathbb{C}$: $s \otimes t = s \otimes (s/s)t = 1 \otimes (s/t)$ so the tensor product is a single copy of $\mathbb{C}$; these rings are not isomorphism as they have different dimension. □

Exercise 2.9. Let G be a finitely generated abelian group of rank r. Prove that $G \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q}^r$.

Proof. We first prove two basic results allowing us to manipulate the group decomposition.

Lemma 1: $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z} \cong \mathbb{Q}$: $\mathbb{Q}$ is a $\mathbb{Z}$-module so tensoring it over $\mathbb{Z}$ is just the module (Example 2.2).

Lemma 2: $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/m\mathbb{Z} \cong 0$: $q \otimes_{\mathbb{Z}} n = q(n/n) \otimes_{\mathbb{Z}} n = q/n \otimes_{\mathbb{Z}} 0 = 0$, so each pure tensor is identically zero.

By the structure theorem for finitely generated abelian groups,

$$G \cong \mathbb{Z}^r \oplus \mathbb{Z}/q_1\mathbb{Z} + \cdots + \mathbb{Z}/q_n\mathbb{Z}$$

Tensoring with $\mathbb{Q}$ gives:

$$\begin{aligned} G \otimes_{\mathbb{Z}} \mathbb{Q} &\cong (\mathbb{Z}^r \oplus \mathbb{Z}/q_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/q_n\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Q} \\ &\cong (\mathbb{Z}^r \otimes_{\mathbb{Z}} \mathbb{Q}) \oplus (\mathbb{Z}/q_1\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}) \oplus \cdots \oplus (\mathbb{Z}/q_n\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}) \\ &\cong (\bigoplus_{i=1}^{r} \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Q} \\ &\cong \mathbb{Q}^r \end{aligned}$$

□

Prove that for infinitely primes p, $G \otimes_{\mathbb{Z}} \mathbb{Z}/p\mathbb{Z} \cong (\mathbb{Z}/p\mathbb{Z})^r$.

Proof. $\mathbb{Z} \otimes_{\mathbb{Z}} (\mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ by Example 2.2 (tensoring an R-module with R is equal to that R-module), and by Exercise 2.2, $(\mathbb{Z}/p\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/q\mathbb{Z}) \cong 0$ for distinct primes p, q.

Therefore for any $p \notin \{q_1, \ldots, q_n\}$ we have:

$$\begin{aligned} G \otimes_{\mathbb{Z}} (\mathbb{Z}/p\mathbb{Z}) &\cong (\mathbb{Z}^r \oplus \mathbb{Z}/q_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/q_n\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/p\mathbb{Z}) \\ &\cong (\mathbb{Z}^r \otimes_{\mathbb{Z}} \mathbb{Z}/p\mathbb{Z}) \oplus (\mathbb{Z}/q_1\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/p\mathbb{Z}) \oplus \cdots \oplus (\mathbb{Z}/q_n\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/p\mathbb{Z}) \\ &\cong (\bigoplus_{i=1}^{r} \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Z}/p\mathbb{Z} \\ &\cong (\mathbb{Z}/p\mathbb{Z})^r \end{aligned}$$

□

Exercise 2.14. Prove that, according to the definition given in §2.4

$$\operatorname{Tor}_0^R(M, N) = M \otimes_R N$$

Proof. Take a resolution of M; then the following is exact:

$$\cdots \longrightarrow R^{\oplus S_1} \xrightarrow{\psi} R^{\oplus S_0} \xrightarrow{\varphi} M \longrightarrow 0$$

We extract an exact sequence from this:

$$0 \longrightarrow \ker \psi \longrightarrow R^{\oplus S_1} \xrightarrow{\psi} R^{\oplus S_0} \xrightarrow{\varphi} M \longrightarrow 0$$

Tensoring with N gives the following exact sequence:

$$\ker \psi \otimes_R N \longrightarrow N^{\oplus S_1} \xrightarrow{1\otimes\psi} N^{\oplus S_0} \xrightarrow{1\otimes\varphi} M \otimes_R N \xrightarrow{0} 0$$

Therefore:

$$\operatorname{Tor}_0^R(M, N) = \ker(0 \circ (1 \otimes \varphi))/\operatorname{img}(1 \otimes \psi) = \operatorname{coker}(1 \otimes \psi) = M \otimes N$$

□

Exercise 2.15. Prove that for $r \in R$ a non-zero divisor and N an R-module, the module $\operatorname{Tor}_1^R(R/(r), N)$ is isomorphic to the r-torsion of N, that is the submodule of elements $n \in N$ such that $rn = 0$. (This is the reason Tor is called Tor).

Proof. Take a resolution of $R/(n)$:

$$0 \longrightarrow R \xrightarrow{r} R \xrightarrow{\sim/(r)} R/(r) \longrightarrow 0$$

Tensoring with N ($R \otimes N = N$) gives an exact sequence:

$$N \xrightarrow{r} N \xrightarrow{\sim/(r)} N/(r)N \longrightarrow 0$$

We extend the exact sequence on the left:

$$\ker(\cdot\, r) \longrightarrow N \xrightarrow{\cdot\, r} N \xrightarrow{\sim/(r)} N/(r)N \longrightarrow 0$$

As this is exact, $a \in \operatorname{Tor}_1^R(R/(r), N)$ if and only if a is in the kernel of multiplying by r; that is, if $ar = 0$. □

Exercise 3.2. If $f : R \to S$ is a ring homomorphism and M, N are S-modules (hence R-modules by restriction of scalars), prove there is a canonical homomorphism of R-modules $M \otimes_R N \to M \otimes_S N$.

Proof. Let $\varphi : M \times N \to M \otimes_S N$ be defined by $\varphi(m, n) = m \otimes_S n$. φ is an R-bilinear map, with multiplication by r interpreted in S by $f(r)$:

$$\varphi(rm, n) = \varphi(f(r)m, n) = f(r)m \otimes_S n = m \otimes_S f(r)n = \varphi(m, f(r)n) = \varphi(m, rn)$$

Therefore there is a unique ψ making the following diagram commute; ψ is our canonical homomorphism between $M \otimes_R N$ and $M \otimes_S N$.

$$\begin{array}{ccc} M \times N & \xrightarrow{\varphi} & M \otimes_S N \\ \downarrow{\scriptstyle \otimes} & \nearrow{\scriptstyle \exists! \psi} & \\ M \otimes_R N & & \end{array}$$

□